Econometrics I

Problem Set 5: Binary variables

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Some binary variable models

1. **CROSS SECTIONAL DATA** Answer the following using the stata datafile `http://www.stata.com/data/s4poe5/data/cps5_small.dta`

   (a) How do average salaries differ between those with and without high-school education? Answer after creating a high-school dummy indicating more than 9 years of education.

   > **Solution:** See **Lecture 9a** and R script**PS5 q1 - wages**: They differ by 64%.

   (b) How do wages increase per year of education, considering that people with and without high school education have a different initial salary, and forcing the same slope?

   > **Solution:** See **Lecture 9a** and R script **PS5 q1 - wages**: The common slope (with different intercepts) is such that an additional year of school education increases wages by 10.8%.

   (c) Does the wage increase per year depend on wether high school was visited?

   > **Solution:** See **Lecture 9a** and R script **PS5 q1 - wages**:

   (d) How does the answer to (a) change if you distinguish between blacks and whites?

Solution: Yes, it depends. Somebody with high school education experiences on average a salary increase of 0.101+0.011 = 11.2% whereas those with no high school education experiences on average a salary increase of just 0.011 = 1.1%. See R script **PS5 q1 - wages**

(e) How does the chance of having high-school education depend on being black?

Solution: The question involves regressing highSchool (a dummy variable) on black (another dummy). As explained during the tutorial, the fitted values correspond to the probability of the outcome highSchool = 1 taking place. Thus, the "slope" is interpreted as the change in the probability that highSchool = 1 happens.

The probability of having high school education is 0.2% lower for black people but that estimate is not significantly different from zero. See first column (1) below and R script **PS5 q1 - wages**.

	Dependent variable:	
	highSchool	
	(1)	(2)
black	−0.002	−0.022
	(0.019)	(0.030)
female		0.036***
		(0.011)
black:female		0.022
		(0.039)
Constant	0.964***	0.949***
	(0.006)	(0.007)
Observations	1,200	1,200
R^2	0.00001	0.010
Note:	*p<0.1; **p<0.05; ***p<0.01	

(f) How does that chance interact with being female?

Solution: Females have a 3.6% higher chance to have high school education. The interaction with black is positive but not statistically significant. See first column (2) above and R script **PS5 q1 - wages**

2. **TIME SERIES DATA**. Answer the following using the stata datafile **passengers.txt**.

(a) Plot the number of passengers vs. time.

Solution: See R code **PS5 q2 - Passengers**.

(b) Estimate a liner time trend of log(passengers) and plot it together with log(passengers) against time.

Solution: See R code **PS5 q2 - Passengers**.

(c) Generate dummies for each of the 12 months. Fit them to the trend residuals and plot them together over time.

Solution: See R code **PS5 q2 - Passengers**.

(d) Regress log(passengers) to both the trend and the dummies. Use that model to make a two-year forecast and plot it.

Solution: See R code **PS5 q2 - Passengers**.

(e) After looking at the regression coefficients, confirm the forecast by hand for November and December in the first forecasted year.

Solution: See R code **PS5 q2 - Passengers**. The approximate estimates are:

- for November: $4.705 + 0.01 \cdot 155 - 0.114 = 6.141$
- for December: $4.705 + 0.01 \cdot 155 = 6.255$

3. **PANEL DATA**. Answer the following using **lifeexpetancyPanel.xlsx**.

(a) Regress life expectancy on log GDP per capita without controlling for fixed effects.

Solution: See **Lecture 9a** and code **PS5 q3 - Life Expetancy**

(b) Now include fixed effects as dummies.

Solution: See **Lecture 9a** and code **PS5 q3 - Life Expetancy**

(c) Now estimate the fixed and random effects models with R's plm library.

Solution: See **Lecture 9a** and code **PS5 q3 - Life Expetancy**

(d) Compare the results.

Solution: See **Lecture 9a** and code **PS5 q3 - Life Expetancy**